1.

List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

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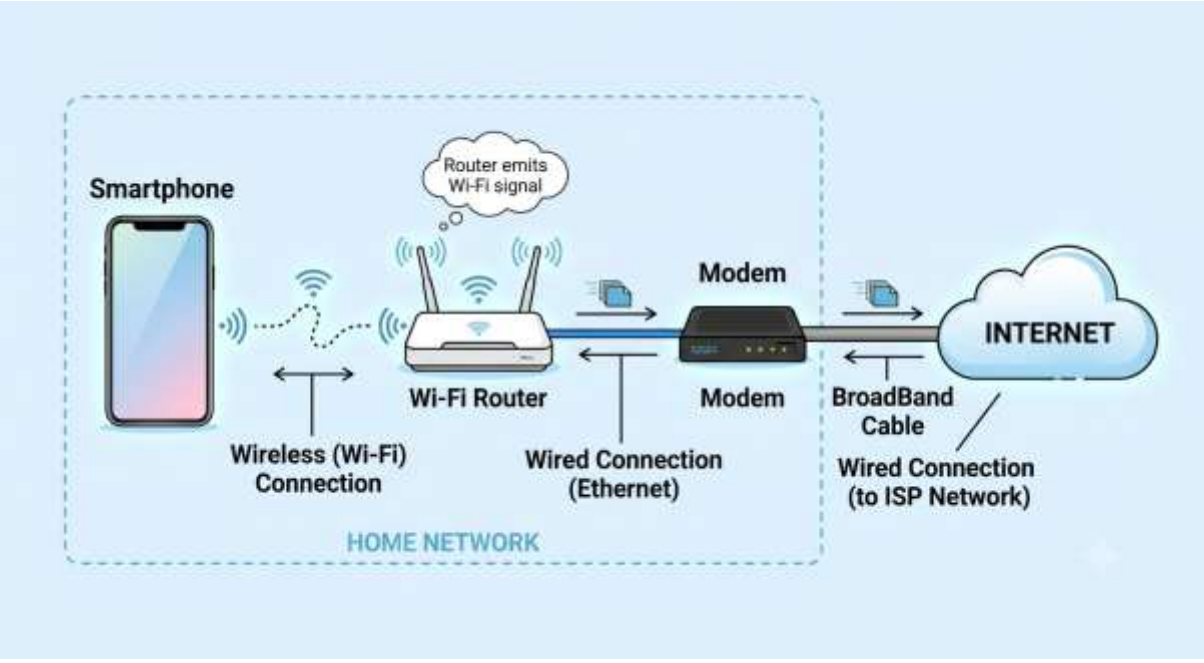
Popular Wi-Fi Standards and Key Differences

Wi-Fi Standard	Common Name	Key Difference in Speed or Frequency Band
802.11n	Wi-Fi 4	Operates on both 2.4 GHz and 5 GHz frequency bands, delivering theoretical speeds up to 600 Mbps.
802.11ac	Wi-Fi 5	Operates exclusively on the 5 GHz band, using wider channel bonding to deliver multi-gigabit speeds starting around 1.3 Gbps.
802.11ax	Wi-Fi 6 / 6E	Expands across 2.4 GHz, 5 GHz, and 6 GHz (Wi-Fi 6E) bands with OFDMA technology for high efficiency in crowded areas, supporting speeds up to 9.6 Gbps.

2.

Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.

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3.

Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why. Hint: Consider both signal strength and security when choosing.

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Wi-Fi Scan Documentation & Network Selection

Sample Wi-Fi Scan Results:

- **Network A (Home_Network_5G)**
    - **Signal Strength:** Excellent (-45 dBm / Full bars)
    - **Security Type:** WPA3-Personal
- **Network B (Public_Cafe_Guest)**
    - **Signal Strength:** Good (-65 dBm / 3 bars)
    - **Security Type:** Open (No Password)
- **Network C (Neighbor_Broadband_2.4G)**
    - **Signal Strength:** Fair (-82 dBm / 2 bars)
    - **Security Type:** WPA2-PSK

Network Choice & Explanation:

- **Selected Network:** Home_Network_5G
- **Why:** This network is the best choice because it combines **optimal signal strength** (preventing connection drops and high latency) with the **highest available security standard (WPA3)**. In contrast, Public_Cafe_Guest offers zero encryption (leaving your traffic vulnerable to snooping), and Neighbor_Broadband_2.4G has a weak signal that would result in slow data rates and unstable connectivity.

4.

Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

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SSID Hiding: Definition, Advantages, and Disadvantages

- **What is SSID Hiding?**

SSID (Service Set Identifier) hiding is a router setting that stops your wireless router from broadcasting its network name to the surrounding area. When enabled, your network name will not show up in the default Wi-Fi drop-down list on nearby devices.

- **Advantage:**

    - **Security by Obscurity from Casual Users:** It deters casual neighbors or passersby from noticing your network name, reducing the likelihood that unauthorized people will attempt to guess your password or spam your connection requests.

- **Disadvantage:**
    - **False Sense of Security & Convenience Loss:** It does *not* provide real cryptographic security. Anyone with basic packet-sniffing tools (like Wireshark) can easily capture the hidden SSID name the moment any device connects to it. Furthermore, it makes connecting your own legitimate devices tedious, as you must manually type out the exact SSID, capitalization, and security type every time a new device joins.